

Running Zoa Waste Hub on Netlify

From the Git repository to a working site at **zoahub.netlify.app**: build settings, database, environment variables, first sign-in, integrations, scheduled jobs and troubleshooting.

Repository: github.com/kefahamis/safisha · Applies to commit 3fb7594 and later · September 2026

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1 Get the code onto `main`

Netlify builds the `main` branch. Merge the latest work from `feature/finance-branding-notifications`, which includes the health report and the database connection fix:

github.com/kefahamis/safisha/compare/main...feature/finance-branding-notifications → **Create pull request** → **Merge**.

2 Check the build settings

In Netlify: **Site configuration** → **Build & deploy** → **Build settings**.

Setting	Value
Base directory	Empty
Build command	<code>npm run build</code> . It migrates the database, then builds the app.
Publish directory	<code>.next</code>
Production branch	<code>main</code>

`netlify.toml` in the repository already sets these, plus Node 22 and the scheduled-functions folder, and it overrides the dashboard. Leave the dashboard fields blank or matching. Netlify adds its Next.js runtime automatically; nothing to install.

3 Create the database

Option A: Netlify DB (simplest)

Extensions → **Neon / Netlify DB** → **Add database**. Netlify sets `NETLIFY_DATABASE_URL` and `NETLIFY_DATABASE_URL_UNPOOLED` for you. Skip `DATABASE_URL` in step 4.

Option B: your own Neon project

In the Neon console under **Connection details**, copy two connection strings:

- the **pooled** one, whose host ends in `-pooler`, for `DATABASE_URL`;
- the **direct** one, for `DATABASE_URL_UNPOOLED`.

Both must end with `?sslmode=require`.

Use one option, not both. When `DATABASE_URL` is set, the app always uses it.

4 Set the environment variables

Site configuration → **Environment variables** → **Add a variable**. For every variable keep **Scopes: All** (at least **Builds** and **Functions**). A variable scoped only to Builds is what makes the health check report `not-configured`.

Generate each secret on your own computer, running this once per secret:

```
node -e "console.log(require('crypto').randomBytes(32).toString('hex'))"
```

Variable	Value	Needed?
<code>SESSION_SECRET</code>	A generated secret	Yes. Sign-in fails without it.
<code>CRON_SECRET</code>	Another generated secret	Yes , for billing and maintenance.
<code>DATABASE_URL</code>	Pooled Neon connection string	Yes , unless you use Netlify DB.
<code>DATABASE_URL_UNPOOLED</code>	Direct Neon connection string	Recommended alongside <code>DATABASE_URL</code> .
<code>DEMO_DATA</code>	<code>1</code>	Only for a demo or test site. Leave unset for real clients.
<code>ADMIN_EMAIL</code> <code>ADMIN_PASSWORD</code>	Your email, and a password of 12+ characters	For a real (non-demo) site. Creates the first platform admin.
<code>BACKUP_ENCRYPTION_KEY</code>	A generated secret. Keep a copy in your password manager.	Turns on nightly backups. They can't be restored without it.
<code>SETTINGS_ENCRYPTION_KEY</code>	A generated secret	Optional. Derived from <code>SESSION_SECRET</code> when unset.
<code>SENTRY_DSN</code> <code>NEXT_PUBLIC_SENTRY_DSN</code>	From your Sentry project	Optional error alerts.

Never set `ALLOW_EMBEDDED_DB` or `ALLOW_DEMO_CODES` on Netlify.

5 Deploy

Deploys → **Trigger deploy** → **Clear cache and deploy site**. Variable changes only take effect after a new deploy.

Near the start of the deploy log you should see:

```
migrate: database is up to date.
```

If it says `migrate: no DATABASE_URL or NETLIFY_DATABASE_URL, skipping`, the database variable isn't reaching builds.

6 Check the site

Open <https://zoahub.netlify.app/api/health>. A healthy site answers:

```
{"ok":true,"db":"up","source":"DATABASE_URL", ...}
```

If it says `"ok":false`, the `reason` tells you what to fix:

reason	Fix
<code>not-configured</code>	No database variable reaches the functions. Check the variable's scopes, then redeploy.
<code>unreachable</code> · <code>ENOTFOUND</code>	Wrong host, or the Neon database was deleted. Copy the connection string again.
<code>unreachable</code> · <code>ECONNREFUSED</code> / <code>ETIMEDOUT</code>	The Neon compute is off or the host is wrong. Check the project in Neon.
<code>login-refused</code>	Wrong username or password in the string. Copy it again, resetting the password if needed.
<code>no-such-database</code>	The database name at the end of the string doesn't exist.
<code>bad-connection-string</code>	Not a full <code>postgresql://...</code> URL, or it has quotes or spaces around it.
<code>setup-failed</code>	Connected, but setup failed. <code>stage</code> and <code>code</code> say where; send them to your developer.

Then open <https://zoahub.netlify.app/>. You should see the public website, not a redirect to `/login`.

7 First sign-in

Demo site (`DEMO_DATA=1`): `/login` lists the demo accounts, all with the password `zoa12345`.

Real site: sign in with `ADMIN_EMAIL` and `ADMIN_PASSWORD`, and set up two-step sign-in when asked. Then:

1. **Admin** → **Companies & estates:** onboard the companies and their estates.
2. **Admin** → **Users:** invite each company's admin.
3. Remove `ADMIN_PASSWORD` from Netlify afterwards. It's only used once.

8 Settings inside the app

As platform admin, open **Admin** → **Settings**:

1. **Public address:** <https://zoahub.netlify.app>, or your custom domain. Safaricom and Africa's Talking send their callbacks here.
2. **SMS (Africa's Talking):** username and API key → **Test connection** → **Text me a test**. Add your phone number to your own account first.
3. **Email (Resend):** API key and a from-address on a domain verified in Resend → **Test connection** → **Email me a test**.

4. **M-Pesa**, per company (the company admin, or you): sandbox keys → **Test connection** → **Register Paybill URLs** → run the Paybill simulator. The README's pilot checklist covers moving to production keys.
5. **Backups** panel, at the bottom of Settings: once `BACKUP_ENCRYPTION_KEY` is set, click **Back up now** and check a backup appears.

9 Scheduled jobs

Under **Logs** → **Functions** you should see two scheduled functions:

Function	Runs	Does
<code>cron-billing</code>	Daily, 08:00 Nairobi	Raises monthly charges and sends payment reminders.
<code>cron-maintenance</code>	Daily, 02:30 Nairobi	Takes the nightly backup, moves photos into Netlify Blobs, and cleans up rate-limit counters.

To check them before waiting a day, open each function under **Logs** → **Functions** and click **Run now**. The log should show `cron maintenance: HTTP 200 ...`.

- `HTTP 401`: the deploy has a different `CRON_SECRET` from the variables. Redeploy.
- `not configured`: `CRON_SECRET` is missing.

10 Monitoring (optional)

- **Uptime**: in UptimeRobot or Better Stack, add a check on `https://zoahub.netlify.app/api/health`. It alerts on anything other than HTTP 200.
- **Errors**: create a Sentry project, put its DSN in `SENTRY_DSN` and `NEXT_PUBLIC_SENTRY_DSN`, and set alert rules in Sentry. Reports carry the error and route only: no cookies, headers or request bodies.

11 Custom domain (optional)

Domain management → **Add a domain**, follow Netlify's DNS steps, and wait for HTTPS. Then update **Public address** in the app (step 8.1) and press **Register Paybill URLs** again, so Safaricom calls the new address.

12 If something's still wrong

Collect these and send them to your developer:

- What `/api/health` returns.
- Any red lines from **Deploys** → **(latest deploy)** → **Deploy log**.
- Any errors under **Logs** → **Functions**.

Good to know

- Netlify caps how long a function can run. The scheduled jobs fit easily at pilot size; watch the maintenance run's duration in the function logs as the data grows.
- Photos and backups use Netlify Blobs, which need no setup and are private to the site.
- A local development server keeps photos in its own database instead; that's expected.

Quick checklist

- ☐ Latest code merged into `main`
- ☐ Database created (Netlify DB, or Neon with pooled and direct strings)
- ☐ `SESSION_SECRET`, `CRON_SECRET` and the database variable set, with Functions in scope
- ☐ `DEMO_DATA=1` for a demo, or `ADMIN_EMAIL` / `ADMIN_PASSWORD` for a real site
- ☐ `BACKUP_ENCRYPTION_KEY` set, with a copy in your password manager
- ☐ Cleared cache and redeployed; deploy log says "database is up to date"
- ☐ `/api/health` returns `"ok":true`; `/` shows the public website
- ☐ Signed in; public address, SMS, email and M-Pesa set up and tested
- ☐ Both scheduled functions run with HTTP 200
- ☐ Uptime monitor on `/api/health`